Paleomagnetism and correlation of some Middle Keweenawan rocks, Lake Superior

H. C. PALMER
Department of Geophysics, The University of Western Ontario, London, Ontario

Received April 13, 1970
Accepted for publication August 12, 1970

Paleomagnetic results from the Lake Superior region appear to indicate that Middle Keweenawan formations or magnetic divisions within formations can be correlated on the basis of magnetic polarity. The Logan sills, the volcanics of the Osler Group, the lava section at Alona Bay, the lower 6000 ft (1829 m) of North Shore volcanics, the lower 1000 ft (305 m) of Cape Gargantua volcanics and the lower 3000 ft (914 m) at Mamainse Point are of reverse polarity. Normal directions of magnetization, which differ from the reverse directions of magnetization by amounts ranging from 140° to 163° occur throughout the Michipicoten Island section, and in the upper parts of the North Shore, Cape Gargantua, and Mamainse Point sections. Provided part of the section at Mamainse Point has been repeated by faulting, these new results, together with data from the literature, are compatible with a single polarity reversal in Middle Keweenawan time separating older reverse polarity rocks from stratigraphically younger normal polarity rocks. The mean direction of magnetization from a conglomerate test is defined with higher precision than would be expected from a randomly distributed population. It is believed that this mean direction reflects a secondary component of magnetization which cannot be preferentially removed by alternating field demagnetization and which has caused the non anti-parallel alignment of normal and reverse populations of magnetization. An estimate of the pole position during Middle Keweenawan time can, however, be made by calculating a pole given equal weight to the mean normal and mean reverse pole. This is calculated to be 159° W, 42° N.

Introduction

The paleomagnetic survey of the Keweenawan rocks in the Lake Superior region by Dubois (1962) illustrated the potential of the paleomagnetic method in supplementing radiometric dating for the purpose of effecting stratigraphic correlations in unfossiliferous rocks. Both remanent magnetic polarity and the pole positions calculated from the directions of remanent magnetization were used to correlate the various Keweenawan sequences around Lake Superior. Some of the Middle Keweenawan volcanic units examined by Dubois (1962), exhibited reverse (positive) magnetic polarity, whereas other units consisted entirely of normal (negative) polarity, but it was only for the Logan sills and dikes that Dubois (1962) had evidence suggesting the age relations between these two sets of magnetic directions.

The efforts of the present study were concentrated on the volcanic sequences which are generally assigned to the Middle Keweenawan (Halls 1966, p. 5). Some of these have not been sampled by previous investigators, while others, such as that at Mamainse Point, have only been partly studied (Dubois 1962). The purpose of this study, therefore, is to obtain a more comprehensive sample of the Keweenawan volcanic sequences, in order to discover if the age relations between normal and reverse populations of magnetization, suggested by the earlier results from the Logan sills and dikes, are generally applicable to the Middle Keweenawan as a whole.

Figure 1 is a generalized geological map of the Lake Superior region showing the main areas of outcrop of Keweenawan volcanic and intrusive rocks and areas sampled for this investigation. These are as follows:

(1) The Logan sills in the Lake Nipigon and Port Arthur (Thunder Bay) regions.
(2) The volcanics of the Osler Group.
(3) The North Shore volcanics of Minnesota.
(4) The section of volcanics at Cape Gargantua.
(5) The section of volcanics on Michipicoten Island.
(6) The section of volcanics at Mamainse Point and Alona Bay.

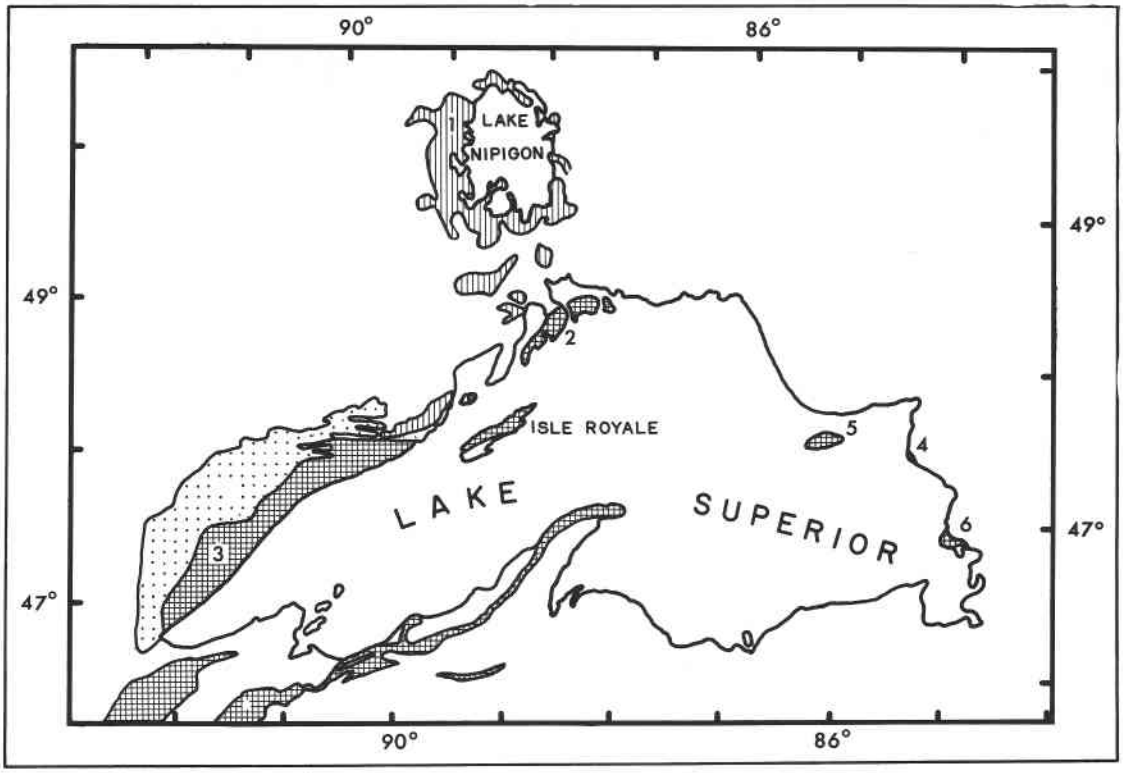


FIG. 1. Index map showing the major regions of outcrop of Middle Keweenawan rocks in the Lake Superior region. Vertical ruling = Logan sills; cross hatching = volcanic sequences; stippling = Duluth Gabbro. Numbers identify areas sampled in this investigation and are identified in the text.

Field and Laboratory Procedures

During most of this study block samples were collected. The blocks were knocked out of outcrops, refitted into position, and then oriented. At some outcrops orientation by means of a Brunton magnetic compass was unreliable and sightings to landmarks of known bearing or sun shots were used as a check of the azimuthal component of orientation. During the last phase of the field work a portable diamond drill was used for sample collecting. Samples were oriented by a device similar to that described by Helsley (1967).

In the laboratory, cores were drilled from each oriented block sample to provide 2 to 4 specimens per sample. Of these specimens, 526 were 3.5 cms in diameter for use on our astatic system and 179 were of 2.5 cms diameter for measurement on a spinner magnetometer. Thirty-five spinner-sized specimens were cut from the field-drilled cores.

In most paleomagnetic work three or more independently oriented samples are collected at each site (a site being defined as a small area within which there are good reasons for supposing that the magnetization was acquired during a time that is short compared with the time scale of secular variation, i.e. an outcrop of a single lava flow). By analyzing within site and between site precision estimates, the magnitude of secular variation may be estimated during the time span represented by the formation (Creer 1962). Because of the great thickness of most of the Middle Keweenawan sections sampled and because of the large number of individual flows in these sections, this type of sampling scheme was not attempted. At most sites only one sample was collected, with the sites distributed as uniformly as possible stratigraphically.

After the initial measurements of NRM (natural remanent magnetization) directions

and intensities, the specimens were subjected to AC demagnetization in steps of 50 Oe rms up to a maximum of 200 Oe rms and remeasured after each of the stages of demagnetization. The specimens from four of the six formations studied were given further AC demagnetization treatment at 300 Oe rms and the Cape Gargantua specimens were taken one stage further to 400 Oe rms. Sample mean directions were calculated after each stage of demagnetization and in cases where two samples were collected from a site, site-mean directions were calculated. Within-sample and within-site precision estimates as given by Fisher's (1953) estimate "k" were calculated and sites or samples for which $k < 20$ throughout the demagnetization treatment were discarded. The number of such discarded sites from each formation is indicated in the appropriate table. The "magnetically cleaned" directions plotted for each formation are generally those measured after the highest stage of AC treatment. For a few of the sites, the within-site or within-sample precision estimate was much higher after a lower stage of demagnetization than at the highest stage. In such cases, the sample direction or site direction measured after this lower AC treatment was used in the calculation of the formation mean direction.

Logan Sills

The Logan sills and dikes were the most comprehensively sampled of all the units studied by Dubois (1962). The sills were resampled by the author in order to obtain an interlaboratory comparison between the earlier results and those of this study. No conscious attempt was made to sample the same sites chosen by Dubois, but samples were collected from the two main regions where outcrops of the sills are abundant, namely near Thunder Bay and around Lake Nipigon. Sample sites are shown in Fig. 2.

The "magnetically cleaned" directions of remanent magnetization are shown in Figs. 3 and 5. All directions from the sills are of reverse polarity. Of all the units sampled in this study the sills in the Lake Nipigon region exhibit the greatest improvement in the precision of the mean direction after AC demagnetization treatment. As an illustration, Figs. 4 and 5 show the sample mean directions for these sills before and after AC demagnetization.

Table I summarizes the paleomagnetic results and in Table IX A the pole position is compared with that calculated by Dubois (1962) for the Logan sills. The differences between the results of this study and those of Dubois (1962) are not great and are primarily one of a difference in inclination. This slight difference may be real, rather than due to orientational or instrumental errors, because Dubois' samples from both the Thunder Bay and Lake Nipigon regions were from more southerly localities than those collected by the author.

Osler Group

The Osler Series was the name given by Tanton (1931) to the succession of flows and sedimentary rocks disconformably overlying the Sibley sedimentary rocks on the peninsula south of Black Bay and on several islands south of Nipigon Bay. The sedimentary members of the group occur predominantly at the base of the section and are followed by a succession of basaltic lava flows, which have an estimated maximum exposed thickness of 3500 ft (1067 m) on Simpson Island. Figure 2 shows the locations of the paleomagnetic sampling sites. Because the lavas are almost flat-lying at the western exposures, the density of sampling sites is not evenly distributed over the stratigraphic sequence; half the sites are in the lower third of the estimated 3000 ft (914 m) sampled section of basalt. All 24 of the samples, which were collected from 12 sites, when measured initially exhibited reverse directions of magnetization, which remained as such after 300 Oe rms AC demagnetization treatment. These results are plotted in Fig. 6 and the statistical data for the lavas of the Osler Group are summarized in Table II. In row 3 of Table II the mean direction of remanent magnetization is calculated without reference to the estimated paleohorizontal (without the structural dip correction). A discussion of the significance of this mean direction and similar mean directions in Tables III, IV, V, and VI is included under the heading 'Fold Test'.

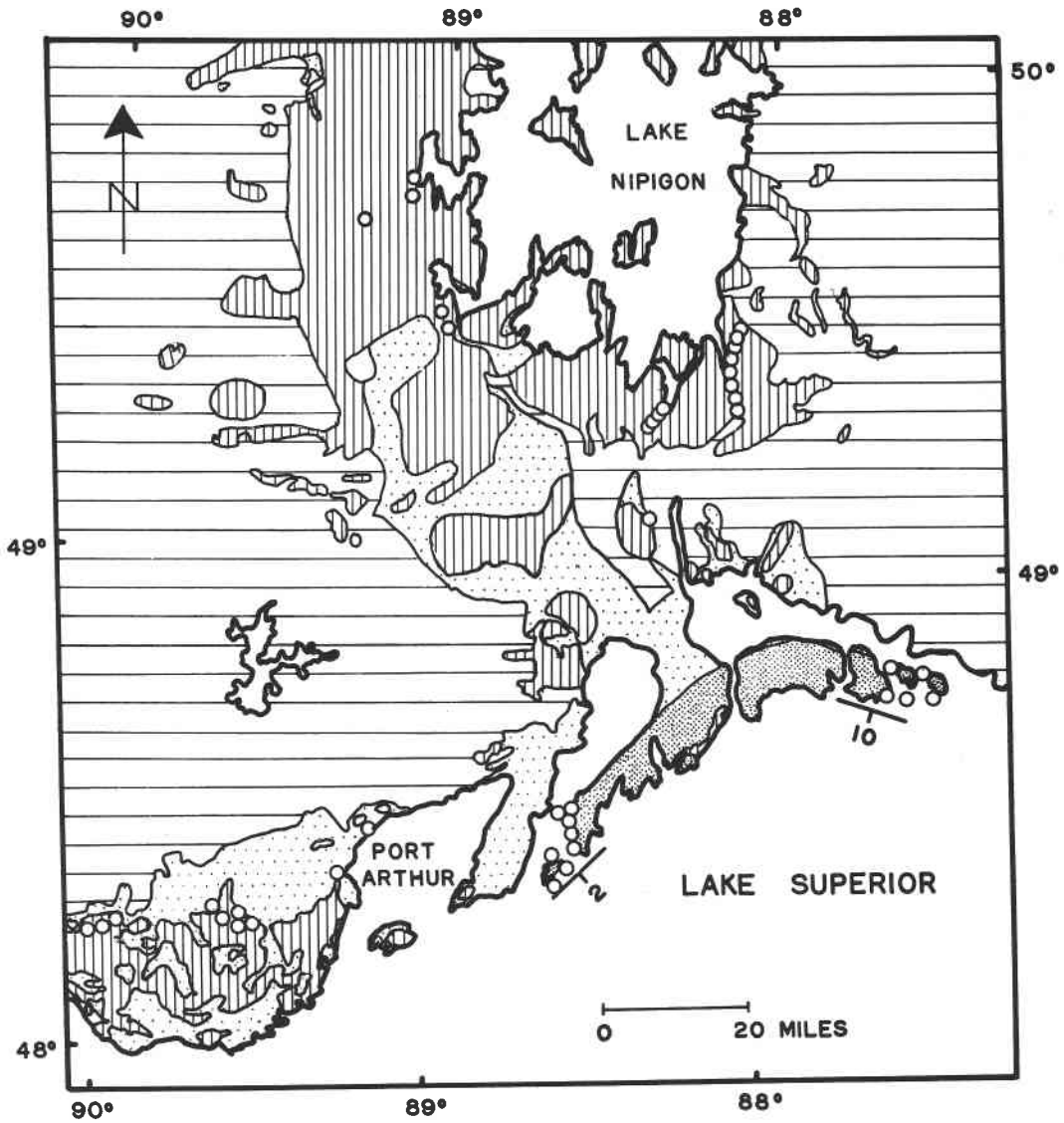


FIG. 2. Generalized geological map of the Port Arthur – Lake Nipigon area showing sampling sites (open circles) in the Logan sills (vertical ruling) and the Osler Group volcanics (close stippling). Horizontal ruling = Archean rocks; open stippling = Post-Archean, pre-Middle Keweenawan rocks. Geology modified from Ont. Dept. Mines Maps 1958B and 2137.

North Shore Volcanics

The North Shore Volcanics is the name given to the very thick sequence of lava flows and minor interbedded sedimentary rocks cropping out along the Lake Superior shoreline from Grand Portage to Duluth (Fig. 7). In the Grand Portage area the base of the sequence rests upon the Lower Keweenawan Puckwunge Formation. From the base of the sequence at Grand Portage to the top of the exposed section at Tofte an estimated 94 flows constituting 17 543 ft (5347 m) of section are exposed

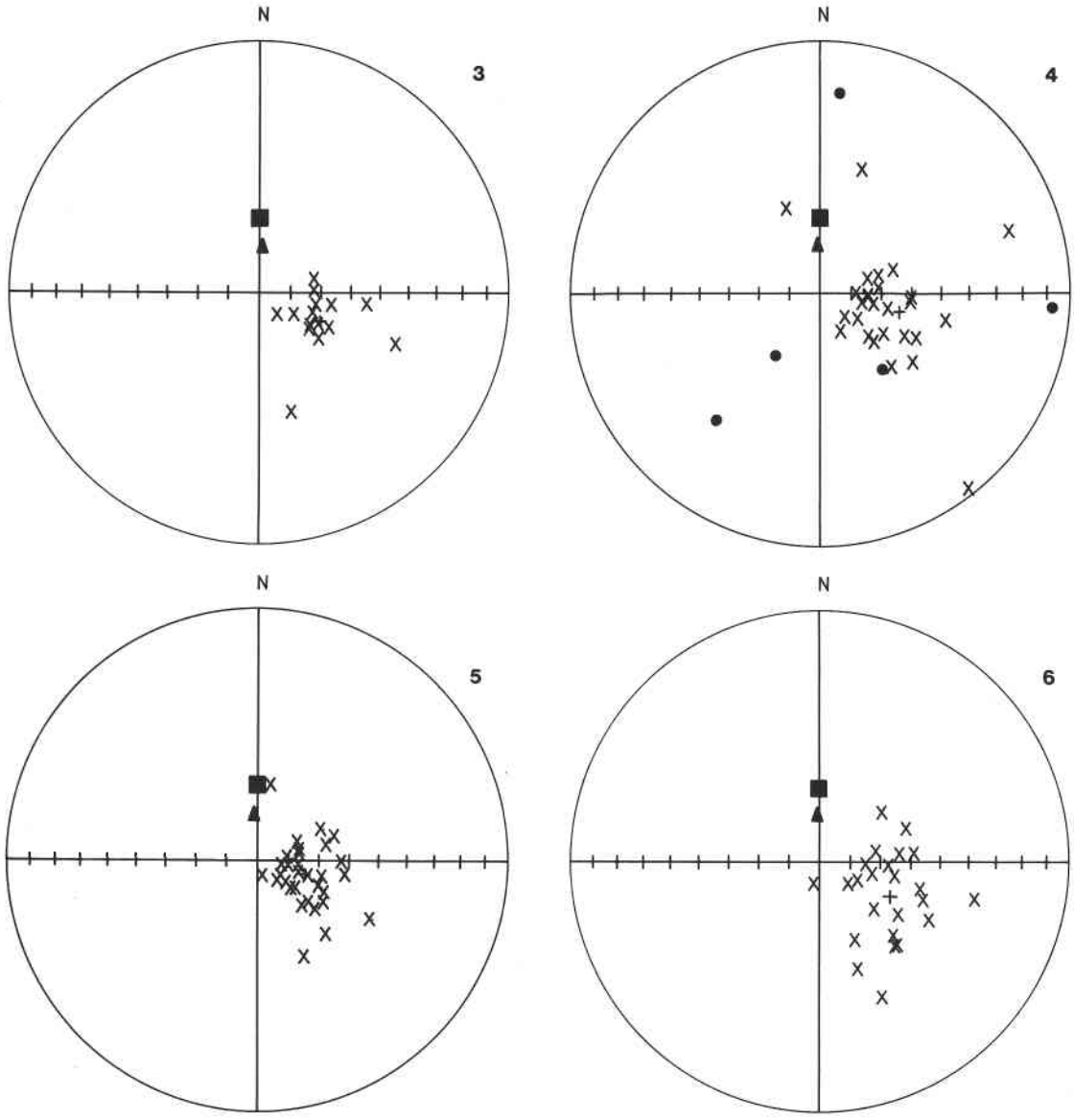


FIG. 3. "Magnetically cleaned" sample mean directions from the Logan sills in the Port Arthur region. Conventions of the equal area projection: N = geographic north; X = north-seeking direction upward; + = mean direction of magnetization; ■ = axial dipole field direction: ▲ = present field direction.

FIG. 4. NRM sample mean directions from the Logan sills in the Lake Nipigon region. Conventions of the equal-area projection: ● = north-seeking direction downward; other symbols are as in Fig. 3.

FIG. 5. Same sample mean directions as plotted in Fig. 4 but after AC demagnetization. Symbol conventions as in Figs. 3 and 4.

FIG. 6. Equal-area projection showing directions of magnetization from samples of the Osler Group volcanics after magnetic cleaning. Symbols as in Fig. 3.

TABLE I

Summary of paleomagnetic data of Logan sills

Locality			N	E	R	D°	I°	k	α95	ϕ^1	λ^1	dp	dm
Lake Nipigon area	1	NRM	21	0	15.195	106.3	−62.0	3.4	20.3°				
" " "	2	"Cleaned"	21	0	20.664	108.3	−73.3	59.4	4.2°	−136.9	49.2	6.7	7.6
Thunder Bay area	3	NRM	9	1	7.863	124.6	−66.7	7.0	20.9°				
" " "	4	"Cleaned"	9	1	8.826	110.7	−69.2	45.8	7.6°	−146.4	47.5	11.2	12.9
" " "	5	Mean pole position from poles in rows 2 and 4								−141.7	48.4		

N = number of site means given unit weight;
E = number of site means excluded in the statistical analysis.
R = resultant of the N unit vectors (Fisher 1953);
D° = declination in degrees east of north;
I° = inclination in degrees (positive downward, negative upward);
k = best estimate of precision (Fisher 1953);
α95 = semi-angle of cone of 95% confidence of mean direction;
ϕ^1,λ^1 = longitude and latitude of the pole position with sign conventions + ϕ^1 = east, + λ^1 = north;
dp, dm = semi-axes of the oval of 95% confidence for the pole position.

TABLE II

Summary of paleomagnetic data of the Osler volcanics
(Sample Localities: 48.4° N, 88.6° W. and 48.7° N, 87.6° W)

Locality	N	E	R	D°	I°	k	α95	ϕ^1	λ^1	dp	dm
1. NRM	12	0	11.153	106.7	−59.0	13.0	12.5°				
2. "Cleaned"	12	0	11.541	117.5	−64.5	24.0	9.0°	−156.7	+48.9	11.7	14.4
3. "Cleaned" (without tilt correction)	12	0	11.498	129.2	−62.4	21.9	9.5°				

NOTE: Symbols as in Table I.

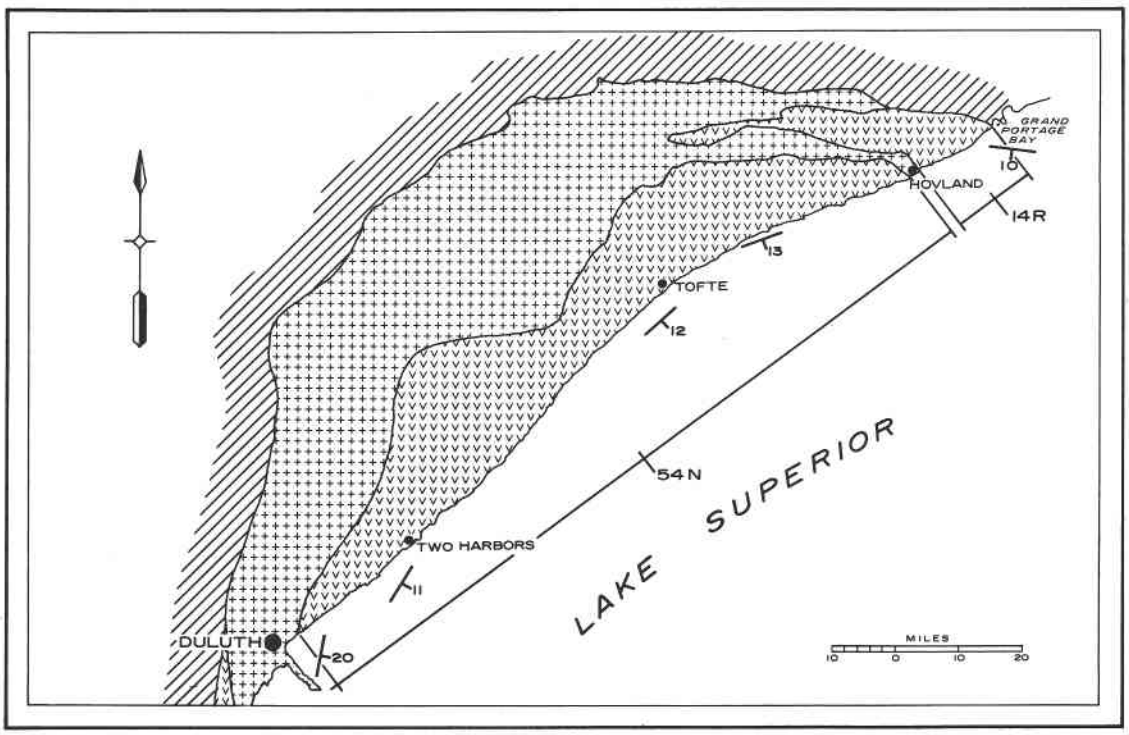


FIG. 7. Generalized geological map of the north shore region of Minnesota modified after Plate 1 of Goldich *et al.* 1961. Diagonal ruling = pre-Middle Keweenawan rocks; + pattern = Duluth Gabbro; V pattern = North Shore volcanics. Brackets off shore indicate numbers of sites of reverse (R) polarity and numbers of sites of normal (N) polarity.

(Grout *et al.* 1959). Drawing largely on the work of Sandberg (1938), Grout *et al.* (1959) estimated that there are 321 flows constituting 25 000 ft (7620 m) of section between Duluth and Tofte.

Seventy-one oriented samples of the volcanics were collected between Grand Portage and Duluth. In addition, 10 samples were collected from the Duluth gabbro, and from diabase sheets and dikes in the Hovland area. Seventeen of the 71 samples are of reverse polarity and the remainder are of normal polarity. The reverse samples are from the lower 27 flows referred to by Grout *et al.* (1959), whereas the normal samples come from flows 34 to 94 plus the unnumbered flows between Tofte and Duluth. Only one of the 10 samples from intrusive rocks is of reverse polarity, and it is from the diabase dike between flows 13 and 14 of Grout *et al.* (1959), which are

TABLE III

Summary of paleomagnetic data of North Shore volcanics

Locality		N	E	R	D°	I°	k	α95	Pole position ϕ^1	λ^1	dp	dm
1. 47.8° N, 89.9° W	NRM	14	3	10.406	127.3	−43.1	3.6	24.4°				
2. "	"Cleaned"	14	3	13.573	116.4	−62.9	30.5	7.3°	−159.8°	47.0°	9.3°	11.7
3. "	"Cleaned" (without tilt correction)	14	3	13.564	134.4	−56.7	29.8	7.4°				
4. 45.0° N, 90.0° W‡	"Cleaned"	11	0		117.0	−59.2	129.4	4.0°	−163.0°	44.5°	(Books 1968)	
5. 47.5° N, 91.0° W	NRM	54	0	50.297	292.1	+52.6	14.3	5.3°				
6. "	"Cleaned"	54	0	51.737	287.7	+47.3	23.4	4.1°	−172.5°	32.1°	3.6°	5.3°
7. "	"Cleaned" (without tilt correction)	54	0	51.507	292.7	+37.0	21.3	4.3°				

NOTE: ‡ = Books (1968) recomputes all measured directions to a common Lake Superior locality of 45 N, 90 W.
All other symbols as in Table I.

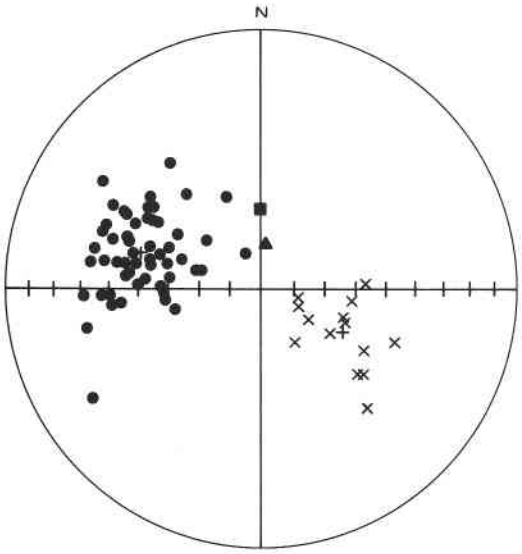


FIG. 8. Equal-area projection showing 14 reverse and 54 normal site mean directions of magnetization after partial AC demagnetization from the North Shore volcanics. Symbol conventions as in Figs. 3 and 4.

within the reverse polarity sequence. The directions of magnetization, after partial AC demagnetization, from the lava flow sites are shown in Fig. 8.

The precision of the mean directions of both the reverse and normal sections increases after AC demagnetization (Table III) and the precision increases for both sections when the directions are corrected for geological dip but this increase is not statistically significant. Recently Books' (1968) has reported results from 11 sites spanning flows 3 to 19 of Grout *et al.* (1959). These results, computed for a common Lake Superior locality of 45° North, 90° West, are given in row 4 of Table III. The computed pole position from Books (1968) data is in good agreement with that of this study.

If the lowermost flows exposed in the Duluth region have not been completely remagnetized by the intrusion of the Duluth Gabbro, which is normally magnetized (Dubois 1962), and if a reversed division (division is used in the rock-stratigraphic sense of Irving and Parry (1963)) was not missed in the sampling from Hovland to Duluth, two conclusions may be drawn: (1) the section of volcanics from Hovland (Flow #27) to the base of the sequence at Grand Portage is older than any of the flows exposed between Tofte and Duluth; (2) the younger 60 to 67 flows exposed from Hovland to Tofte are stratigraphic correlatives with the much thicker sequence of 321 flows from Tofte to Duluth.

Gargantua Cape Volcanics

Ayres (1969) has recently estimated that a cumulative thickness of 2700 ft (823 m) of Middle Keweenawan basalt, conglomerate, and rhyolite overlies pre-Keweenawan rocks in the Cape Gargantua region. The section can conveniently be divided into three basalt members (each member consisting of a number of individual flows averaging 20 to 30 ft (6 to 9 m) in thickness), separated by two conglomerate members, with the top of the exposed section capped by rhyolite (Ayres 1969). Figure 9, adapted from Ayres (1961), shows the distribution of sample sites.

The NRM directions fall into two populations. Sites from the lower half of the section (within the lower basalt member) have reverse directions of magnetization, whereas sites from the upper half of the section yield normal directions of magnetization. The directions of magnetization after 400 Oe rms AC field treatment from the 25 samples collected at Cape Gargantua are plotted in Fig. 10, and the mean directions are recorded in Table IV. Of all the Middle Keweenawan sections studied, the Cape Gargantua section displays the greatest departure from being a 180° reversal. Because of this observation these samples were demagnetized to the higher field of 400 Oe rms in an effort to detect any secondary component of magnetization which may have served to bias the population directions from the more commonly observed 180° reversal pattern. No significant change in direction was found; further discussion of the significance of asymmetric reversals in these Keweenawan sections is given in a later section.

Michipicoten Island Volcanics

Sampling of the volcanic rocks on Michipicoten Island was confined largely to the southern and western shorelines with the sketch map of Burwash (1905) as a guide to the geology. Although neither the base nor the top of the volcanic sequence is exposed, the visible

TABLE IV

Summary of paleomagnetic data from Cape Gargantua

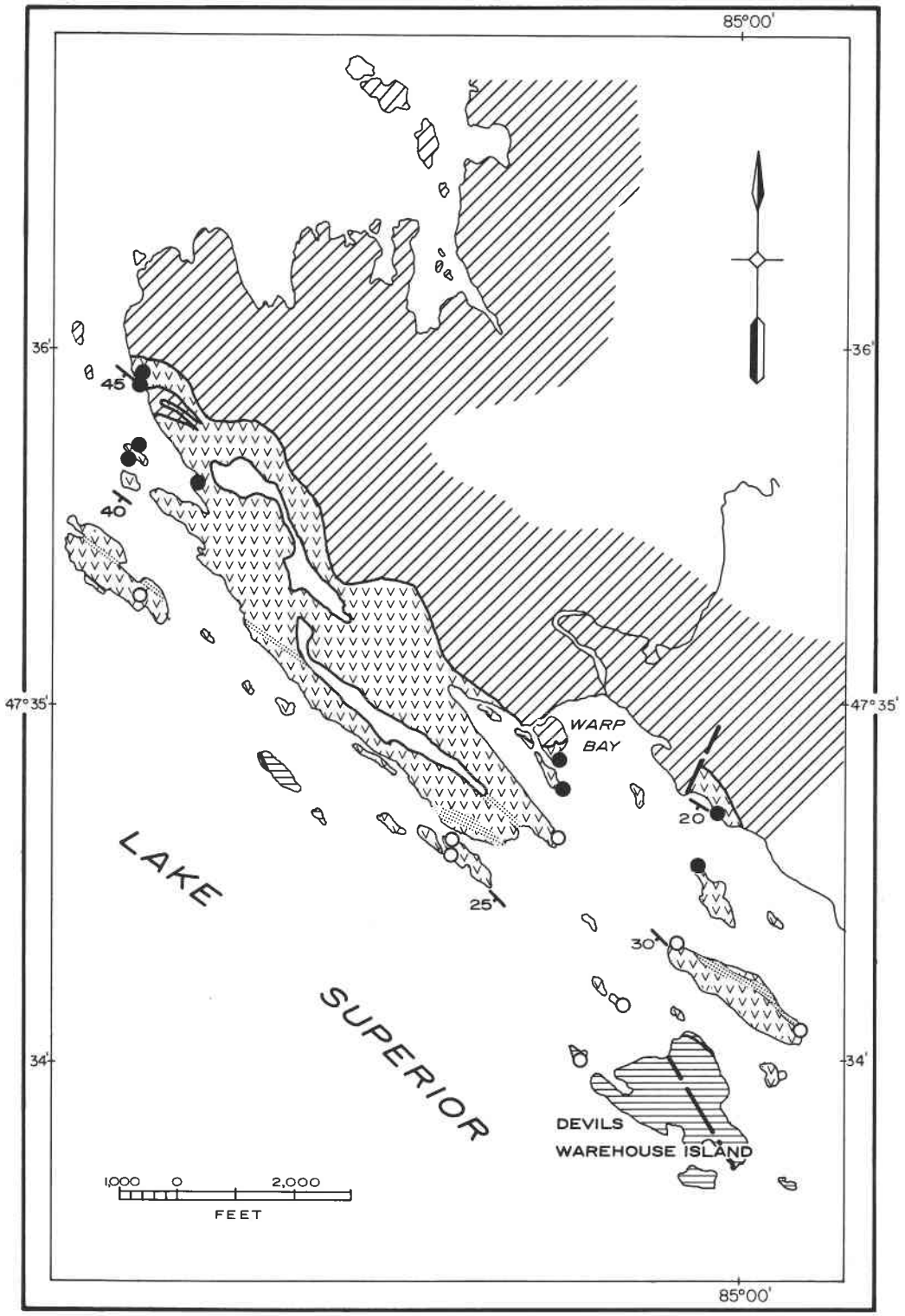
		N	E	R	D°	I°	k	α95	Pole position ϕ^1	Pole position λ^1	Pole position dp	Pole position dm
Upper Normal Division												
47.6° N, 85.0° W	NRM	8	0	7.870	295.9	+40.9	53.8	7.6°				
" "	"Cleaned"	8	0	7.880	297.6	+37.7	58.3	7.3°	+179.7°	+33.9°	5.5°	8.6°
" "	"Cleaned" (without dip correction)	8	0	7.825	319.3	+37.5	40.0	8.9°				
Lower Reverse Division												
47.6° N, 85.0° W	NRM	9	0	8.890	157.5	−72.0	72.5	6.1°				
" "	"Cleaned"	9	0	8.911	154.3	−71.5	89.8	5.5°	−136.7°	+72.1°	8.4°	9.5°
" "	"Cleaned" (without dip correction)	9	0	8.680	189.3	−45.9	25.0	10.5°				

TABLE V

Summary of paleomagnetic results of Michipicoten Island volcanics

Locality		N	E	R	D°	I°	k	α95	Pole position ϕ^1	Pole position λ^1	Pole position dp	Pole position dm
1. 47.7° N, 85.8° W	NRM	28	2	25.896	288.4	+39.4	12.8	7.9°				
2. "	Cleaned	28	2	25.981	293.6	+32.3	13.4	7.7°	+179.0	28.7	5.2	8.7
3. "	Cleaned (without tilt correction)	28	2	25.727	303.8	+16.3	11.9	8.2°				

NOTE: symbols as in Table I.


85°00'
36'
45
40
47°35'
WARP
BAY
20
LAKE
25
30
SUPERIOR
34'
DEVILS
WAREHOUSE ISLAND
1,000 0 2,000
FEET
85°00'

succession is about 11 360 ft (3453 m) thick (Annells 1970).

Forty-three oriented samples from 30 sites were collected; only 28 sites, however, yielded acceptable within-site precision estimates. The locations of these sites are shown in Fig. 11 and the directions of magnetization, all of which are of normal polarity, are plotted in Fig. 12. The statistical data summarizing the paleomagnetic results from the Michipicoten Island lavas are given in Table V. Although the pole position from the Michipicoten Island lavas is in good agreement with the pole positions from other Middle Keweenawan sections of normal polarity, it is defined with low precision. A possible reason for the overall lower precision may be that the values used for dip correction were in some cases inaccurate as they had to be interpolated from values measured lower or higher in the section.

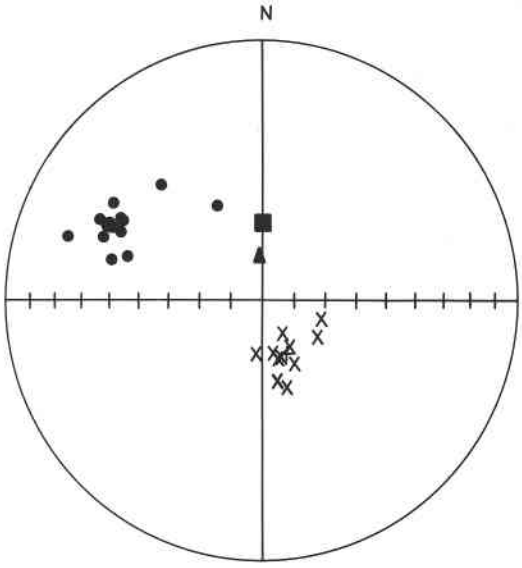


FIG. 10. Sample mean directions after partial AC demagnetization from the Cape Gargantua section of volcanics. Reverse directions (X's) come from sites that stratigraphically underlie sites of normal polarity (●'s).

Alona Bay and Mamainse Point Sections

The Keweenawan rocks at Mamainse Point and Alona Bay have attracted the attention of geologists for many years. Logan's (1863) preliminary account was sufficiently interesting to encourage Macfarlane (1866) to make a more detailed examination of the Mamainse Point "upper copper bearing series". Macfarlane's (1866) estimate of 16 208 ft (4940 m) of interbedded trap and conglomerate are little different from more recent estimates of total thickness of the Middle Keweenawan at Mamainse Point.

Dubois (1962) reported paleomagnetic results from basalt at Alona Bay and from the upper part of the section at Mamainse Point. The Alona Bay samples were of reverse polarity, whereas the Mamainse Point samples were of normal polarity; the pole positions were significantly different, which suggested that the two groups of rocks were of different age. This result is somewhat surprising inasmuch as the basalt at Alona Bay can be considered as an outlier of the much thicker sequence at Mamainse Point. For this reason the thick sequence at Mamainse Point was sampled in some detail, particularly the lower part of the section which had not been sampled previously by Dubois (1962).

Twelve samples from twelve sites were collected from the lavas at Alona Bay using the geological map of Nuffield (1955) as a guide to outcrop localities. Eleven of the twelve samples yielded internally consistent directions of remanent magnetization and these directions after AC cleaning are plotted in Fig. 13 and the mean direction, before and after cleaning (with and without structural dip correction applied), are recorded in Table VI.

The decrease in precision of the mean direction referenced to the estimated paleohorizontal as compared to the precision of the mean without the application of structural dip correction does not necessarily mean that a post-tilting secondary component of magnetization is recorded by these lavas. At most localities the structural attitude was difficult to determine and for some of the sites estimates had to be made from the attitude recorded at adjacent

FIG. 9. Geological map of the Cape Gargantua region adapted from Ayres (1961). Diagonal ruling = Pre-Keweenawan rocks; V pattern = Middle Keweenawan basalt flows; stippling = conglomerate; horizontal ruling = rhyolite. Sampling sites are coded as to polarity: ● = sites of reverse polarity; ○ = sites of normal polarity.

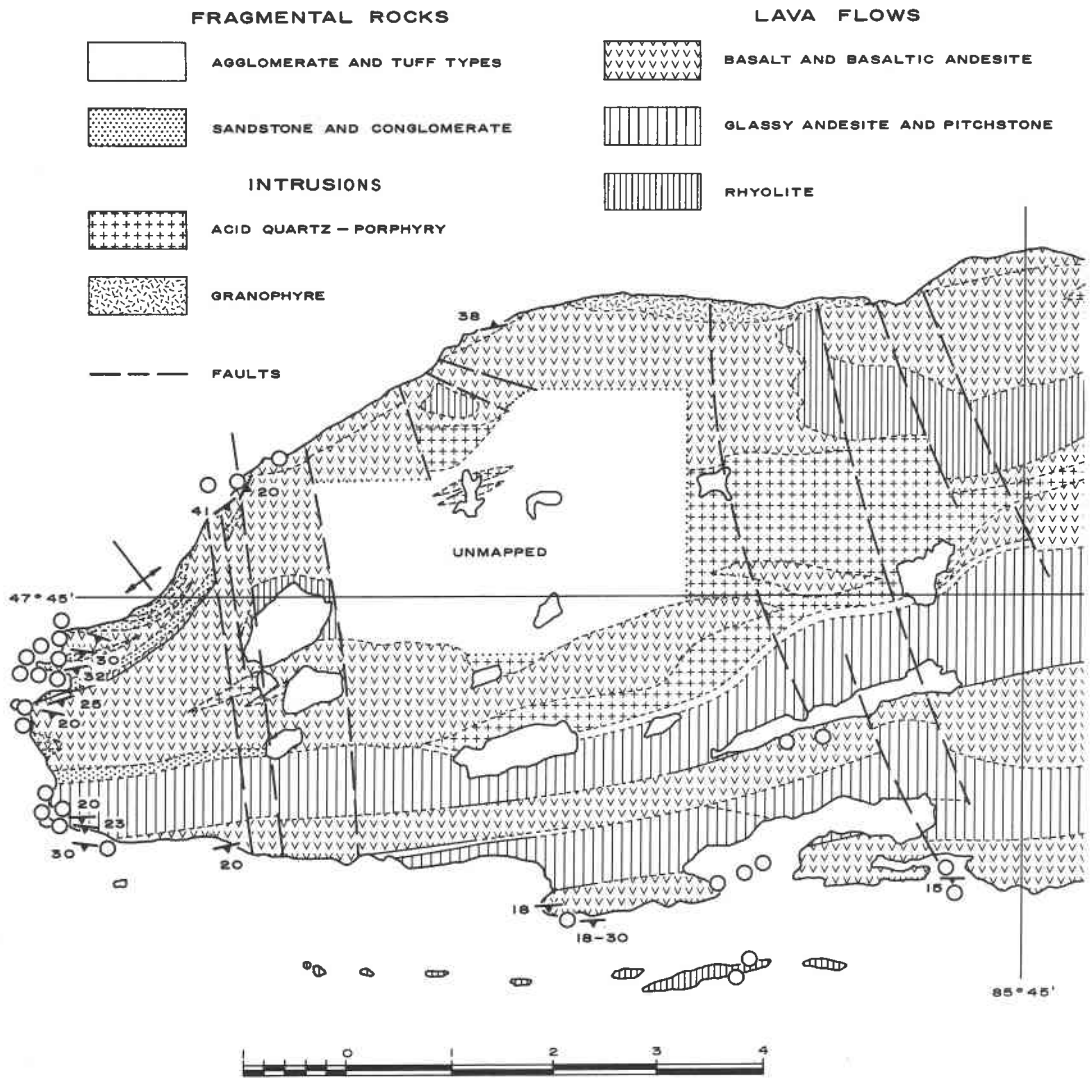


FIG. 11. Geological map of the western half of Michipicoten Island, adapted from a map supplied by R. N. Annells. Sampling sites are indicated by open circles.

sites. Furthermore some of the sites are close to the surface of unconformity separating the lavas from the underlying Archean granite and because this surface was one of considerable relief, initial dips of the lowest flows probably vary appreciably. For both of these reasons the precision and more importantly the accuracy of the Alona Bay paleomagnetic result are adversely affected.

Seventy-nine sites were sampled at Mamainse Point using the map of Thomson (1953) as a guide to outcrop localities for the upper part of the section. Sites east of Thomson's (1953) map were located on air photographs. All sampling sites were replotted on the recently published geological maps of Giblin (1969*a*, *b*, *c*, *d*), which provide geological map coverage for the entire area underlain by Middle Keweenawan in the Mamainse Point area. Most of the samples were collected along the shoreline and along the highway which is parallel to the shoreline and an effort was made to distribute the sample sites as uniformly as possible stratigraphically. A few samples were taken

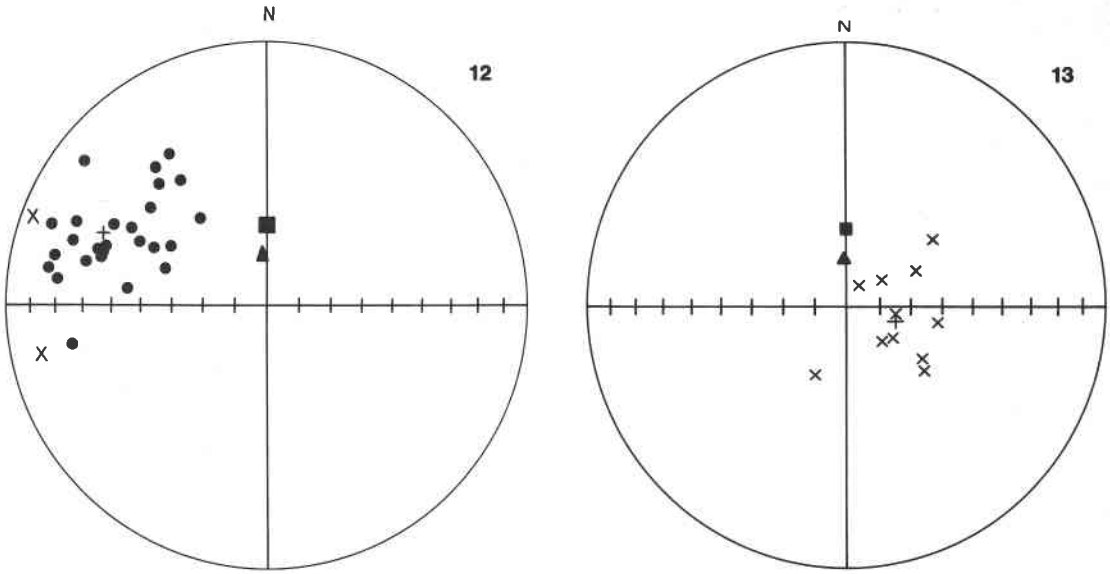


FIG. 12. Site mean directions of magnetization from the Michipicoten Island section of volcanics. Conventions of the equal area projection are as in Figs. 3 and 4.

FIG. 13. Magnetically cleaned, site-mean directions from lavas at Alona Bay. Symbol conventions as in Fig. 3.

from inland sites to provide checks on the stratigraphic continuity of the magnetic divisions defined by the shoreline and highway sites.

The simple pattern of a single polarity division or of a single normal polarity division overlying a single reverse magnetic division observed in the other Middle Keweenawan sections breaks down at Mamainse Point. In common with the other sections where the base of the Middle Keweenawan succession rests unconformably on pre-Middle Keweenawan rocks, the lowest lavas are of reverse polarity. However, rather than being succeeded stratigraphically by a single normal polarity division, the magnetic sequence from the base to the uppermost exposed lavas is: reverse, normal, reverse, normal. The boundaries between these divisions and the number of sites defining these divisions are indicated on Fig. 14. The directions of magnetization of the two lower divisions after partial AC demagnetization to a maximum of 300 Oe rms, are shown in Fig. 15 and the directions of magnetization of the two upper divisions are shown in Fig. 16. The mean directions for each of these divisions, together with their associated statistical parameters, are recorded in Table VI. On Figs. 15 and 16 are shown four directions of magnetization obtained from sandstone lenses within the two prominent conglomerate members shown on Fig. 14.

The transition from reverse magnetic polarity to normal magnetic polarity occurs near the base of each of these conglomerate members. Close-spaced sampling revealed that the transition from reverse to normal polarity occurs in the first exposure of lava immediately beneath each of the two conglomerates. Covered intervals of approximately 12 and 20 ft (4 and 6 m) respectively obscure the contact between lava and conglomerate. This coincidence suggests that the boundary between the lower normal division and the upper reverse division is a strike fault which has repeated the upper part of the lower reverse division and the lower part of the upper normal division. This implies that the two conglomerates shown on Fig. 14 are stratigraphic correlatives with the western conglomerate, being the down-dip, thickened equivalent of the eastern conglomerate. The eastern conglomerate, where examined in the highway road cuts, is composed entirely of Keweenawan basaltic lava clasts whereas the

TABLE VI

Summary of paleomagnetic data from Mamainse Point and Alona Bay

Mamainse Point			N	E	*R*	D°	I°	*k*	α95	ϕ^1	λ^1	dp	dm
Upper normal division	NRM	1	38	3	35.459	301.5	+40.5	14.6	6.3°				
	"Cleaned"	2	38	3	36.245	303.4	+40.0	21.1	5.2°	+177.3°	+39.0°	4.0°	6.2°
	"Cleaned" (without dip correction)	3	38	3	35.847	328.4	+53.4	17.2	5.8°				
Upper reverse division	NRM	4	5	2	4.847	105.1	−58.8	26.2	15.2°				
	"Cleaned"	5	5	2	4.877	106.6	−56.8	32.5	13.6°	−156.7°	+36.8°	15.0°	20.0°
	"Cleaned" (without dip correction)	6	5	2	4.875	170.3	−77.7	32.1	13.7°				
Lower normal division	NRM	7	14	0	13.613	286.8	+41.7	33.6	7.0°				
	"Cleaned"	8	14	0	13.500	290.4	+41.1	26.0	7.9°	−172.2°	+30.7°	6.1°	9.6°
	"Cleaned" (without dip correction)	9	14	0	13.450	323.1	+64.4	23.6	8.4°				
Lower reverse division	NRM	10	15	2	12.634	138.7	−70.6	5.9	17.2°				
	"Cleaned"	11	15	2	14.352	135.0	−70.9	21.6	8.4°	−141.4°	+61.2°	12.6°	14.4°
	"Cleaned" (without dip correction)	12	15	2	14.384	222.5	−57.3	22.7	8.2°				
Normal division (upper & lower normal combined	"Cleaned"	13	52	3	49.599	299.9	+40.4	21.2	4.4°	−179.6°	+36.8°	3.5°	5.3°
	"Cleaned" (without dip correction)	14	52	3	49.105	327.2	+56.4	17.6	4.8°				
Reverse division (upper & lower reverse combined)	"Cleaned"	15	20	4	19.040	124.8	−67.9	19.8	7.5°	−147.6°	+54.4°	10.6°	12.8°
	"Cleaned" (without dip correction)	16	20	4	18.866	216.9	−63.4	16.8	8.6°				
Alona Bay													
Reverse division	NRM	17	11	1	10.274	124.4	−74.6	13.8	12.8°				
	"Cleaned"	18	11	1	10.524	106.3	−74.8	21.0	10.2°	−127.3°	+47.4°	17.1°	18.5°
	"Cleaned" (without dip correction)	19	11	1	10.658	286.8	−73.0	29.2	8.6°				

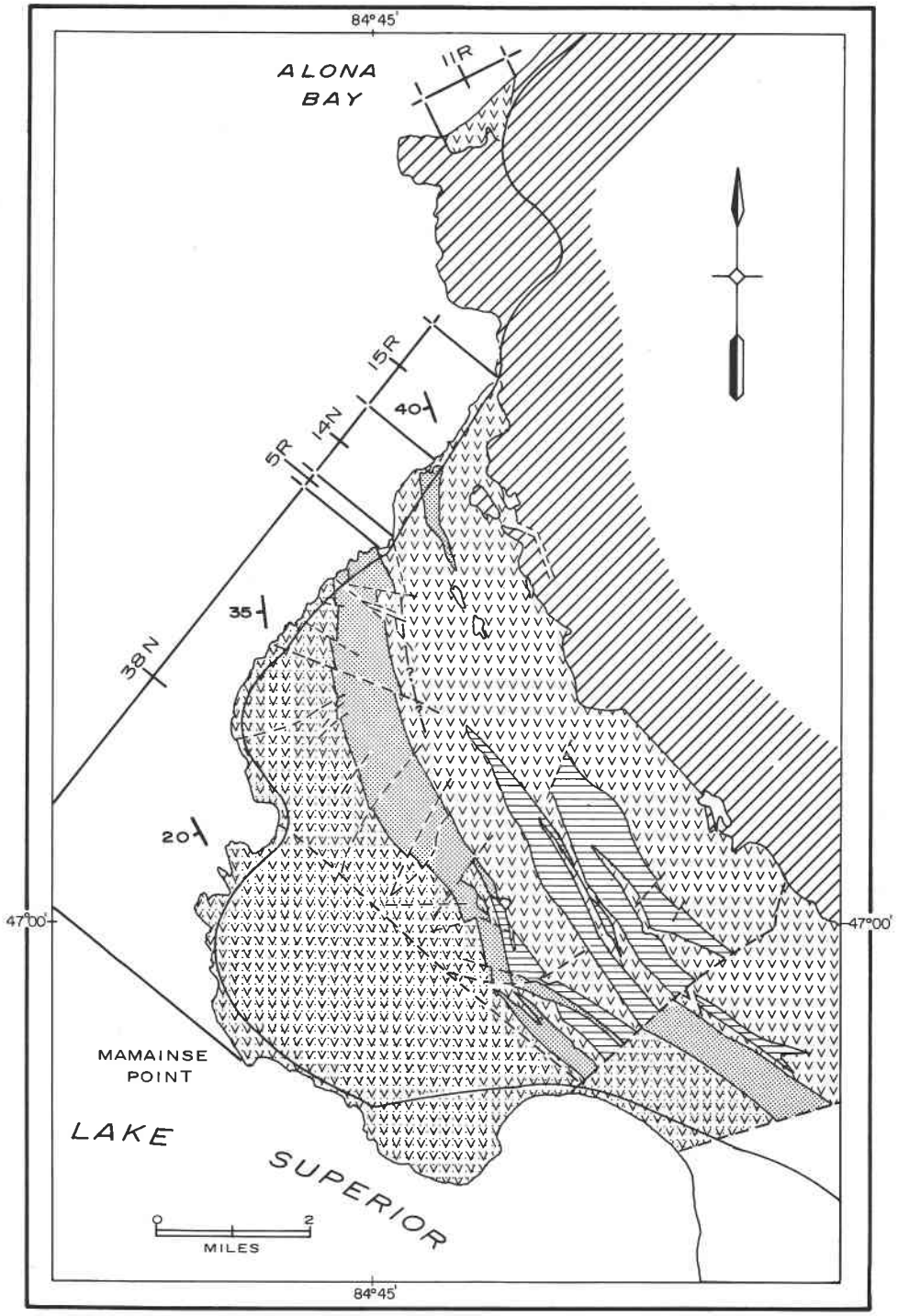


FIG. 14. Geological map of the Mamainse Point and Alona Bay areas (after Giblin and Leahy 1965, and Giblin 1969*a*, *b*, *c*, *d*). Diagonal ruling = pre-Keweenawan rocks. Other patterns show distribution of Middle Keweenawan rocks as follows: V pattern = basic lava flows; horizontal ruling = rhyolite; stippling = conglomerate; mixed V pattern and stippling = basic lava, rhyolite, and conglomerate undivided. Off-shore brackets indicate numbers of sites defining each polarity division. Strike fault east of thicker conglomerate member is inferred from magnetic data.

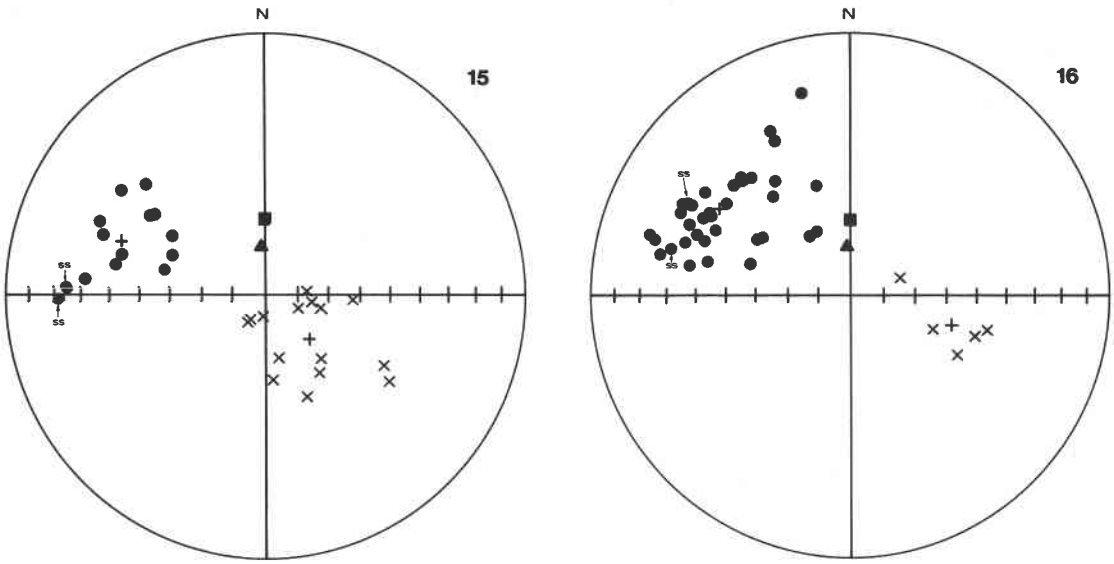


FIG. 15. Magnetically cleaned, site mean directions from the lower two polarity divisions depicted on Fig. 14. Site directions labelled "SS" are directions in sandstone lenses within the lower conglomerate mapped on Fig. 14. Other symbol conventions are as in Figs. 3 and 4.

FIG. 16. Magnetically cleaned, site mean directions from the upper two polarity divisions depicted in Fig. 14. Site directions labelled "SS" are directions in sandstone lenses within the upper conglomerate mapped on Fig. 14.

western one has many Archean granite clasts in addition to Keweenawan lava pebbles and cobbles.

Although there is as yet no geological evidence to support the fault hypothesis, the magnetic evidence favoring repetition by faulting is tentatively accepted, and all normal directions and all reversed directions have been combined to yield a single normal mean direction and a single reverse mean direction as recorded in rows 13 and 15 of Table VI.

Stability Tests

From cutting relationships between the Logan dikes and sills, Dubois (1962) was able to show that the normal directions of magnetization are younger than the reverse directions of magnetization in the Logan intrusives. The same relationship is preserved in the volcanic sequences examined in this study; normal directions of magnetization are associated with rocks that overlie reversely magnetized flows. This raises the possibility that the same reversal has been recorded in all of these volcanic-intrusive Middle Keweenawan formations. This idea would be enhanced if it could be demonstrated that the asymmetry observed between the normal and reverse mean directions is primary and accurately measures the field directions at the time the lavas cooled. In none of the formations are the mean directions 180° apart. The closest approach to a 180° reversal is in the North Shore volcanics, where the mean directions are 163° apart; the greatest departure from a 180° reversal is recorded in the Cape Gargantua lavas, where the mean directions differ by 140°. The essential fact is that the asymmetry is in the same sense in all the formations studied, i.e., the reverse directions are more steeply inclined upward than the normal directions are inclined downward. If the directions can be shown to be primary, then a shift in the dipole field direction accompanied the reversal in polarity. This situation would be reminiscent of the Paterson transition (Irving 1966), which marks the lower boundary of the Late Paleozoic Kiaman polarity interval.

Because reversals which involve a 180° change in direction are by far the most frequent type found in younger geological formations it becomes of extreme importance, when asym-

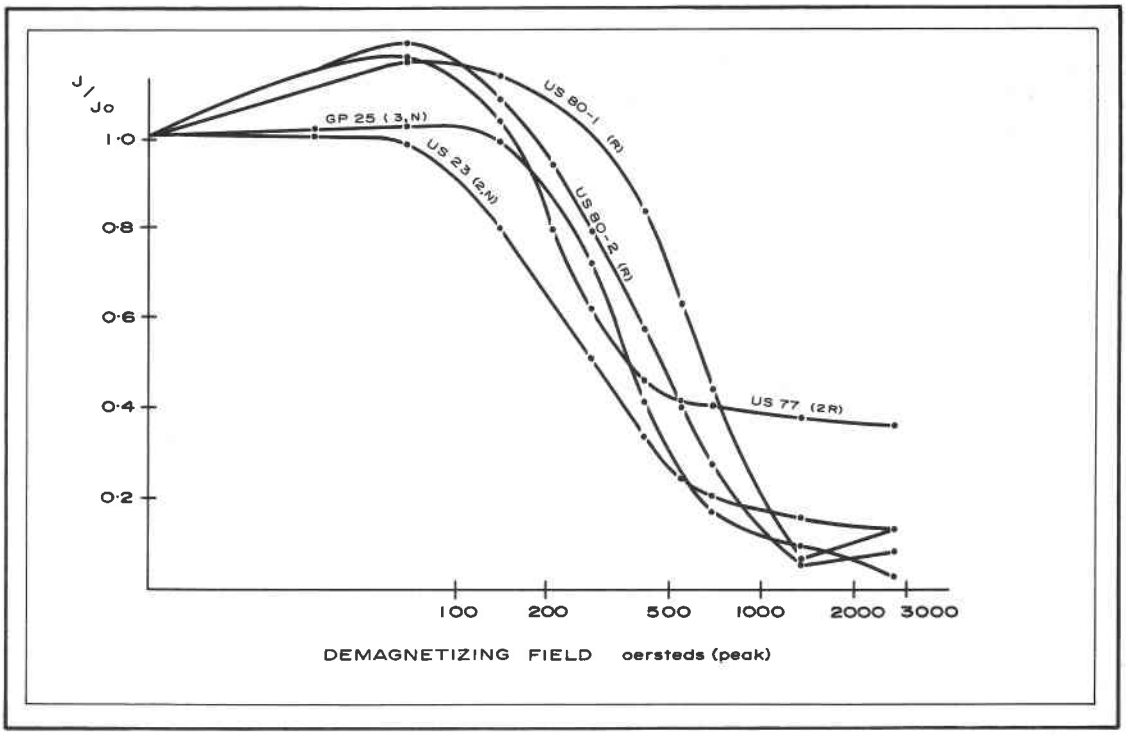


FIG. 17. Alternating field demagnetization (normalized) curves of selected samples used in high AC field demagnetization experiment. Numbers in brackets indicate the number of specimen values used to calculate the average demagnetization curve for the sample; N and R indicate polarity. For the less stable sample US-80, each specimen curve is plotted.

metrically reversed populations of magnetization are found in a formation, to perform experiments which will rule out the possibility that the asymmetry is the result of a secondary component of magnetization. Should evidence be found for a secondary component of magnetization then methods to remove it should be sought or alternatively if an estimate can be made of its magnitude in relation to the primary magnetization, the measured directions can be corrected to give an estimate of the original direction. With these possibilities in mind, the following experiments were performed.

(1) High Field Demagnetization Studies

A number of normally and reversely magnetized samples were selected from the collection of previously cored block samples and additional virgin cores were drilled from them providing 2 or 3 specimens per independently oriented sample. Each of the specimens was subjected to AC demagnetization at 35, 70, 140, 280, 420, 560, 700 Oe (peak) and the direction and intensity of remanent magnetization remeasured after each stage of demagnetization. The experiment was completed at the paleomagnetic laboratory at the Southwest Center for Advanced Studies, Dallas, where equipment for performing AC demagnetization to high fields was put at the disposal of the author. The specimens were redemagnetized at 700 Oe peak, remeasured, and further demagnetized at two higher stages, i.e. 1400 Oe and 2800 Oe peak.

Figure 17 shows typical demagnetization curves obtained in this experiment. The majority of these samples are magnetically hard and stable, as is evidenced by the respectable grouping of the initial NRM directions plotted in Fig. 18.

Eleven of the sixteen samples remained stably magnetized up to the maximum applied field of 2800 Oe. In the remaining samples,

the precision of the sample mean remained high up to some lower field treatment, but with further treatment scattering of the directions set in. Specimens 1 and 2 of sample US-80 (Fig. 17) are examples of these less stably magnetized samples. In this sample, and in two others similar to it, the direction obtained after the 700 Oe (peak) treatment was chosen as the cleaned direction. For the remaining two samples, the directions after the 560 Oe treatment and the 1400 Oe treatment respectively were used. The directions obtained have been plotted in Fig. 19. When compared with Fig. 18, it can be seen that the precision has increased in each of the two populations, but the population means have not moved towards a more antiparallel distribution; the asymmetry of the reversal still remains.

The results of this experiment indicate that the postulated secondary component, if present, is not a viscous magnetization or partial thermoremanent magnetization (PTRM) (Briden 1965), as these types of secondary magnetization are destroyed in moderate AC fields. However, many of the lava flows show the effects of burial metamorphism, and secondary hematite resulting from the breakdown of primary ferromagnesian silicates and similar to the hematite illustrated in Figs. 20 and 21 of Wilson and Haggerty (1966), is common in many of the flows. Should this hematite carry a remanence it would be of a chemical remanent magnetization (CRM) type. Chemical remanence particularly in hematite, would not be preferentially removed with respect to the original TRM by the AC demagnetization method.

(2) Fold Test

For all formations studied with the exception of the Logan sills the present attitude of the lava flows is not the attitude which they had at the time of eruption. For 8 of the 9 polarity divisions, the precision of the mean direction of magnetization is increased when the directions are referenced to the estimated paleohorizontal, but in only one of these polarity divisions is the increase in precision of the mean statistically significant (Tables II–VI). This is to be expected, however, inasmuch as each of the polarity divisions was measured in a homoclinal sequence.

Because the structural attitude differs markedly from area to area a fold test can be applied by comparing the mean directions for each polarity division in the areas studied, calculated with and without structural dip corrections (Fig. 20). The mean direction of the 4 normal polarity division mean directions is defined with a precision estimate "k" (Fisher 1953) of 16.5, which increases to a "k" of 120 when the directions of magnetization are referenced to the paleohorizontal. For the reverse polarity divisions "k" increases from 9.8 (present attitude) to 107 when restored to estimated paleohorizontal. These changes are significant at the 95% and 99% confidence limits (McElhinny 1964).

Most current models on the evolution of the Lake Superior Syncline depict some basinward tilting of the lava flows very soon after eruption. However, the Late Keweenawan Freda and Nonesuch Formations have lakeward dips up to 20° (Dubois 1962, p. 13). The folding continued well into Late Keweenawan time and was not completed with certainty until the time of deposition of the Munising Formation of Late Cambrian age. Thus, the result from the fold test only indicates that the magnetization is pre Late Cambrian in age.

(3) Conglomerate Test

The Middle Keweenawan section at Mamainse Point provides ideal material for the application of the Conglomerate test for magnetic stability (Graham 1949). Essentially, if it can be shown that a parent formation has a good grouping of magnetic directions, whereas, clasts of the parent formation deposited in an overlying formation have randomly scattered directions, then the magnetization pre-dates conglomerate deposition. The Mamainse Point section is ideal in that all the conglomerate layers have unambiguously recognizable Keweenawan lava clasts, and since the conglomerate layers are interbedded with the lava flows, a positive result from the test would prove that the magnetization was acquired at the time of initial cooling of the flows.

Accordingly, 37 independently oriented samples of lava clasts were collected from 2 conglomerate layers in the Mamainse Point section. One of the conglomerates sampled is the eastern one shown on Fig. 14; the second is one of the numerous thin conglomerate beds

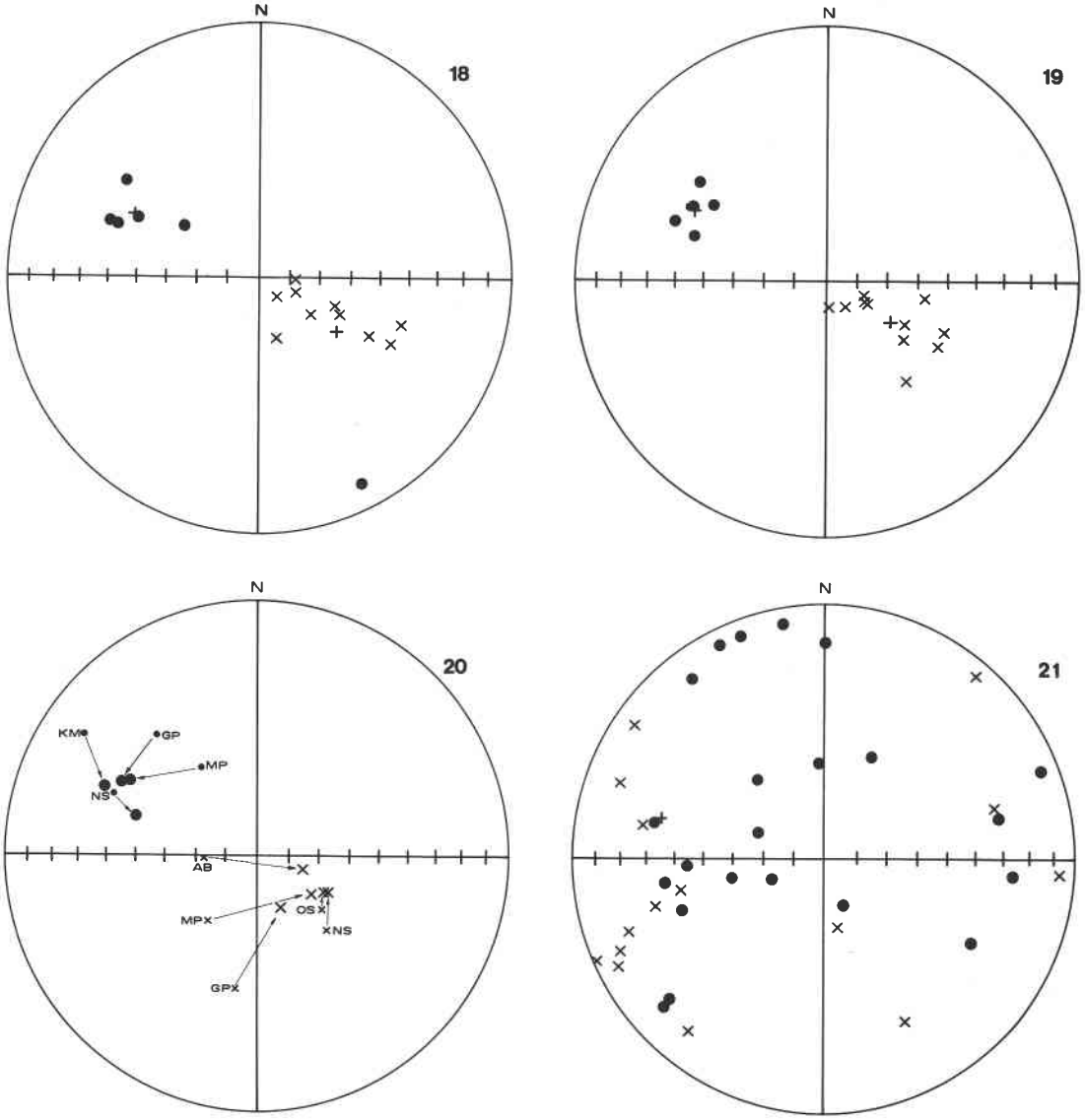


FIG. 18. Initial NRM directions of magnetization of the samples used in the high AC field demagnetization experiment. Symbol conventions as in Figs. 3 and 4.

FIG. 19. Same sample directions as plotted in Fig. 18, but after AC demagnetization in high fields.

FIG. 20. Mean directions of magnetization for each polarity division calculated with directions referenced to present horizontal (small symbols) and to estimated paleohorizontal (large symbols). Arrows point to the polarity division means calculated with the structural dip correction applied. Formations identified as follows: MP, Mamainse Point; KM, Michipicoten Island; GP, Cape Gargantua; NS, North Shore; AB, Alona Bay; OS, Osler volcanics.

FIG. 21. Directions of remanent magnetization, after partial AC demagnetization, in oriented cobbles and boulders of Keweenawan lava taken from two conglomerate layers in the Mamainse Point sequence. Symbols as in Figs. 3 and 4.

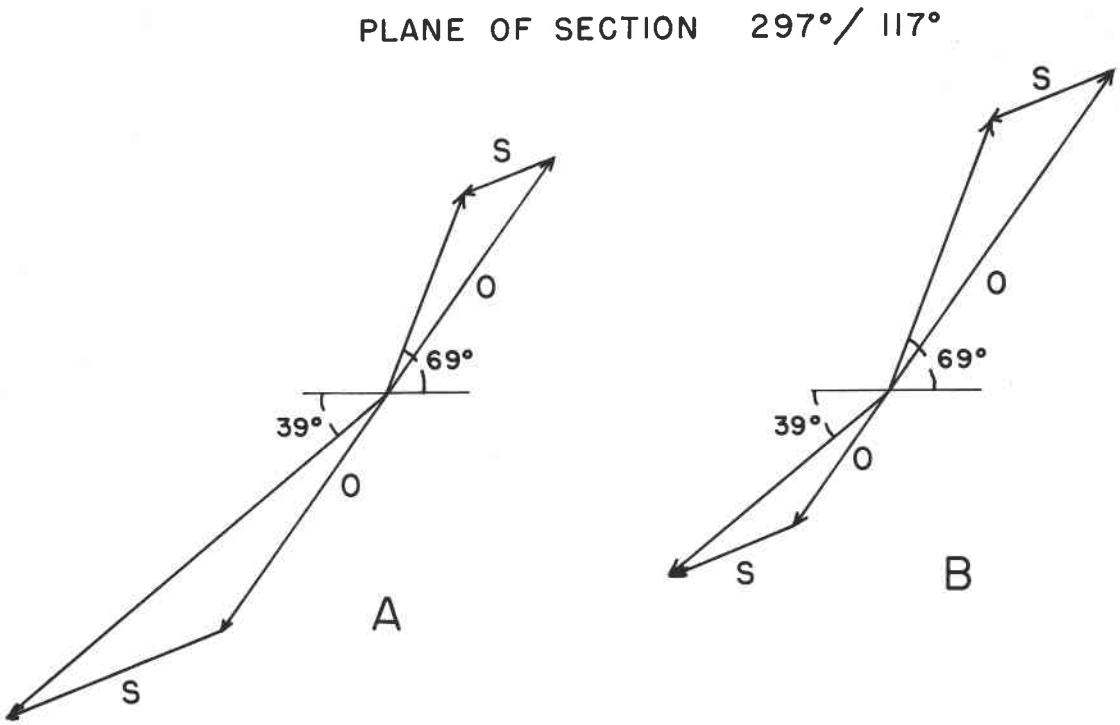


FIG. 22. Vector diagrams indicating how original anti-parallel magnetizations (O) could yield the observed asymmetrically reversed directions by the addition of a secondary component of magnetization (S) in the direction of the mean from the conglomerate test.

within the upper normal polarity division. The directions of magnetization in the conglomerate cobbles were measured and the samples were then subjected to AC demagnetization in the same fashion as were samples from the lava flows. As can be seen from Fig. 21, where these directions have been plotted, the degree of scatter is much higher than from the lava flows. This result gives one considerable confidence that a substantial proportion of the remanence measured in the lava flows is primary. The statement is qualified because the directions are not uniformly distributed over the sphere; there are more directions in the 3rd and 4th quadrants than elsewhere, and the majority of directions are directed downward. Dubois (1962) performed a similar experiment on lava pebbles collected from the Copper Harbor conglomerate in Michigan with a result similar to that obtained in this study.

The directions of remanent magnetization in the lava cobbles of the Copper Harbor conglomerate given in Fig. 8 of Dubois (1962) were measured on his stereonet plot and used to calculate the statistical parameters given in Table VII where the same information is recorded for the Mamainse Point experiment.

From Table 4.4 of Irving (1964) one can conclude that there is much less than 5% probability that this sample of 74 directions is drawn from a population of uniformly distributed unit vectors. Assuming that the mean direction from the combined conglomerate tests is the best estimate of the direction of a hypothetical post-conglomerate secondary component of magnetization, Fig. 22 was constructed.

TABLE VII

Summary of paleomagnetic data of Mamainse Point and Copper Harbor Conglomerate tests

	N	E	*R*	D°	I°
Mamainse Point	37	0	11.60	283.8	+34.5
Copper Harbor	37	0	12.92	311.0	+09.0
Mamainse Point and Copper Harbor combined	74	0	23.35	299.4	+21.6

The plane of section for Fig. 22 is the average declination of all normal and reversely magnetized Middle Keweenawan formations of this study. In Fig. 22A the original directions are assumed to be equal in magnitude and anti-parallel. The addition of the secondary component to intersect the observed mean directions predicts that the normal intensity of magnetization should be double the reverse intensity of magnetization. This is not observed. For the Mamainse Point lavas the mean intensities after 200 Oe rms cleaning are as follows: normal lavas, 12.4×10^{-4} emu/cm^3 with a standard deviation of 19.0×10^{-4} emu/cm^3; reverse lavas, 12.8×10^{-4} emu/cm^3 with a standard deviation of 20.4×10^{-4} emu/cm^3. In Fig. 22B the observed directions are plotted equal in magnitude and the secondary direction is subtracted to intersect the hypothetical anti-parallel vectors. This model predicts that the original intensities of reverse directions were greater than the original normal intensities. Assuming that equal numbers of reversely magnetized and normally magnetized cobbles were deposited in the conglomerates, the average cobble should have a secondary component of magnetization equal to approximately 50% of the original magnetization. The question then arises as to whether the degree of clustering observed in the conglomerate test could be produced by the superposition of a common secondary component on an original random (uniform) distribution where the magnitude of the secondary component is about 50% of the original.

To obtain an estimate of the magnitude of the secondary component in relation to the primary component, the following experiment was performed. A sample of 74 directions was drawn using a random number generator from a population of 40 962 directions which are spaced uniformly over a sphere. A description of the production of this population of directions will be given at a later date (Starkey, in preparation). The sample of 74 directions was tested for randomness by the use of the statistics of Fisher (1953). Each of the random directions in the sample was then averaged vectorially with a common direction with unit weight given to each. The common direction used was the residual mean of the conglomerate test. The mean direction and the length of the resultant vector R of the 74 resulting directions were calculated. Each random direction was then averaged again with the common direction but with a weight of 2 given to the random direction and a weight of 1 to the common direction. The mean direction and R of the 74 new directions were recalculated. The experiment was continued using ratios of the random direction to the common direction (primary to secondary) of 3:1, 4:1 . . . 10:1.

In total, 22 trials of the experiment were performed and with no common bias applied the values of the length of the resultant vector, R, ranged between 0.5 and 15.4 for the samples of 74 directions. The mean directions were randomly distributed. When the common bias was applied to each random direction in the ratio 1:1, R ranged between 44.0 and 53.0. When the common bias was applied to each random direction in the ratio 1:2, the values of R ranged between 13.9 and 30.8. In the trials where the common direction was given a weight of 1 while the random direction was given a weight of 3, the values of R ranged between 8.5 and 22.8. The value of R in the Mamainse Point and Copper Harbor conglomerate tests combined is 23.35 (Table VII). This value falls within the range which is produced when a secondary component in a specified direction, equal in magnitude to 50% of the original, is superimposed on the original direction.

The results from the stability test experiments are taken to indicate that a secondary component of magnetization is present in these rocks which cannot be successfully removed by AC demagnetization, which predates at least in large part, the folding, and which is sufficient in magnitude to create the observed asymmetry between normal and reversed populations of magnetization.

Age of the Magnetization

A considerable amount of radiometric data is available bearing on the age of some of the Middle Keweenawan formations which have been studied paleomagnetically. In the preceding section evidence was presented indicating that the remanent magnetization, at least in part, was acquired during initial cooling. Because the asymmetry in the opposed polarities has been shown to be the result of a secondary

TABLE VIII

K–Ar ages of the Logan sills and dikes

Sample No.	Location	Calculated age	Reference or analyst
PA-3	48°14′ N, 89°59′ W	1030 ± 40	T. J. Ulrych
PA-16	48°16′ N, 89°35′ W	1060 ± 40	T. J. Ulrych
NP-6	49°26′ N, 88°07′ W	865 ± 60	N. J. Snelling
NP-12	49°19′ N, 88°07′ W	730 ± 50	N. J. Snelling
G.S.C. 61-138	48°01′ N, 89°30′ W	1000 ± 60	Lowdon *et al.* 1963

NOTES: $\lambda\varepsilon = 0.585 \times 10^{-10}$/yr. $\lambda\beta = 4.72 \times 10^{-10}$/yr.

component of magnetization an accurate estimate can be made of the original field direction. Thus if the radiometric dates are true geological ages, the Middle Keweenawan pole position can be assigned an age.

Goldich *et al.* (1961) have reported a number of K–Ar mineral ages ranging from 1000 m.y. to 1200 m.y. for the Duluth Gabbro and county rocks metamorphosed by the gabbro. Four whole-rock samples of granophyre from the Duluth Gabbro yielded K–Ar ages ranging from 750 m.y. to 1030 m.y. (Goldich *et al.* 1966, p. 5404). More precise estimates of the age of the Duluth Gabbro have been provided by the Rb/Sr isochron method and by the U/Pb method. Faure *et al.* (1969) reported a Rb/Sr isochron age of 1115 ± 14 m.y. for the Duluth Gabbro. Silver and Green (1963) found that zircons extracted from a rhyolite flow in the North Shore volcanics and from granophyres in the Duluth Gabbro and the Endion sill, plotted as a linear array on a U/Pb 'Concordia' diagram sharing a collective projected age of 1115 ± 15 m.y.

Four of the paleomagnetic samples from the Logan sills were analyzed for K and ^{40}Ar in the geochronology laboratory of Oxford University. The ages calculated from these results are tabulated in Table VIII, together with one published result from metamorphic biotite produced at the contact of a Logan dike (G.S.C. 61-138). NP-6 and NP-12 are coarser grained than PA-3 and PA-16 and their younger ages when compared to PA-3 and PA-16 probably reflect argon loss. Because of the possibility of varying amounts of argon loss from all samples the results are interpreted as minimum estimates of the time of intrusion of the Logan sills and dikes.

Recent Rb/Sr work on rhyolites from the Mamainse Point sequence gives a reliable estimate of the age of the Middle Keweenawan in that area. It is not certain whether the rhyolites at Mamainse Point are flows or hypabyssal intrusives, but, the occurrence of rhyolite clasts in the interbedded conglomerates (P. E. Giblin, personal communication, 1969) indicates that the rhyolites are contemporaneous with the basalt flows that yielded the paleomagnetic results. W. R. VanSchmus (personal communication, 1970) reports that six rhyolite samples yield a Rb/Sr isochron of 1076 m.y. The analytical uncertainty calculated at the 95% confidence limits is approximately 25 m.y.

In summary, the available radiometric data do not reveal any age difference between rocks of reverse magnetic polarity and rocks of normal magnetic polarity. The Rb/Sr and U/Pb ages of the Duluth Gabbro, the North Shore volcanics, and the Mamainse Point volcanics are equivalent. The age of Middle Keweenawan volcanism and intrusive activity and the concomitant acquisition of thermoremanent magnetization is taken to be about 1100 m.y.

Pole Positions

All Middle Keweenawan pole positions from this and previous studies have been summarized in Table IX. From these data mean pole positions for the reverse and normal polarity divisions have been calculated and are plotted, with their 95% circles of confidence, in Fig. 23. The larger scatter of the reverse polarity poles apparent in Fig. 23 and Table IX was recently noted by Beck and Lindsley (1969); the additional data supplied by this study serves only to re-emphasize their observation. It seems unlikely that the greater scatter of the reverse polarity poles can be attributed entirely to incomplete averaging of paleosecular variation in the reverse polarity divisions studied. If this were the case the precision estimates of the

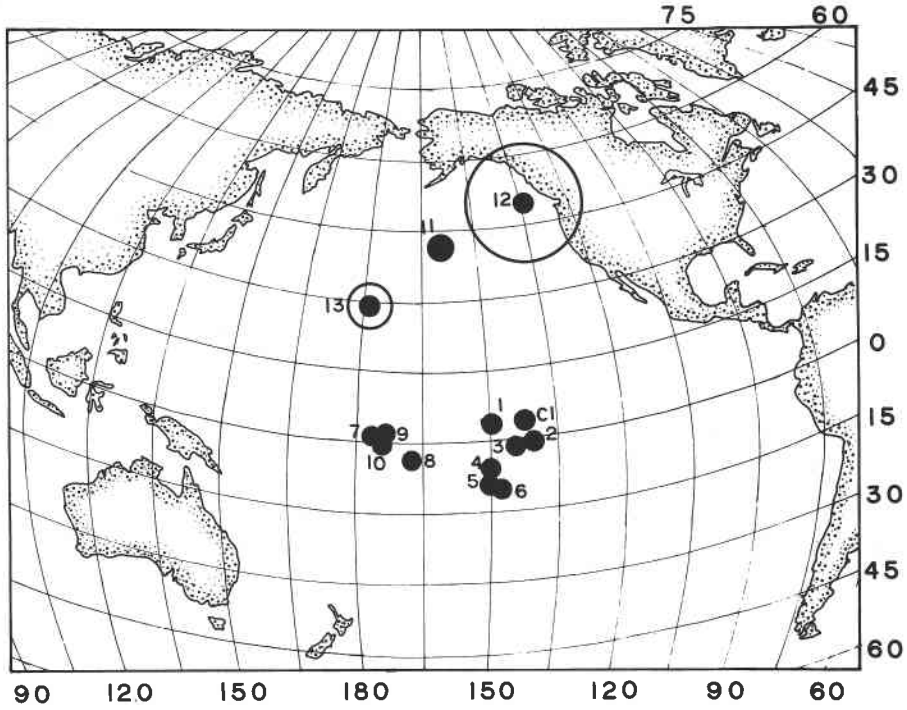


FIG. 23. Ninety-five percent confidence limits are drawn about the mean Middle Keweenawan reverse (12) and normal polarity (13) poles. Pole 11 is the mean of poles 12 and 13. Other poles are explained in text and in Palmer (1969).

mean of each reverse polarity sequence should be systematically greater than precision estimates for the mean of the overlying normal polarity division. This is not observed at Mamainse Point nor in the Ironwood and Portage Lake lava of Michigan (Books 1968, Table I).

The contrast in precision between the reverse polarity and normal polarity poles can more readily be explained by the results of the conglomerate test. Because the secondary direction of magnetization, as inferred from the mean in the conglomerate test, forms a large angle with the original reverse direction but forms only a small angle with respect to the original normal direction the original reverse direction is more sensitive to change in inclination resulting from small differences in the proportion of secondary to primary components of magnetization from formation to formation. Original differences in declination of the reverse polarity divisions have been accentuated because the planes of remagnetization are great circles.

Inasmuch as the asymmetry between normal and reverse populations of magnetization has been shown to have resulted from the acquisition of a secondary component of magnetization, poles 12 and 13 of Fig. 23 are considered invalid. An accurate estimate of the pole position during Middle Keweenawan time can be made by averaging pole positions 12 and 13 (Table IX). This pole is plotted as Pole 11 on Fig. 23.

In Fig. 23 the Middle Keweenawan pole position (pole 11) is compared with those from slightly older rock units of North America. Poles CI and 1 to 10 are from widely separated rock units in North America, ranging in estimated age from 1475 m.y. to 1100 m.y. References to the paleomagnetic data and to the geochronological data for these pole positions and age assignments are given in Palmer (1969). The difference between the pole position given by the Middle Keweenawan rocks of Lake Superior and those supplied by the Belt Series of Montana and the Purcell System of Alberta (poles 5 and 6 of Fig. 23) is great,

TABLE IX

Summary of Middle Keweenawan pole positions

Formation or magnetic division	Reference	Paleomagnetic pole	Mean pole
A. Formations and Magnetic Divisions of Reverse Polarity			
Logan sills	This study	−141.7+48.4	−136.2+51.1
Logan sills I	DuBois (1962)	−130.0+53.5	
Osler Volcanic Group	This study	−156.7+48.9	−156.7+48.9
North Shore volcanics	This study	−159.8+47.0	−161.4+45.8
North Shore volcanics (Grand Portage)	Books (1968)	−163.0+44.5	
Baraga County, dikes	Graham (1953)	−99.5+45.0	−99.5+45.0
Ironwood area, Michigan Lavas	Books (1968)	−127.5+28.6	−127.5+28.6
Cape Gargantua lavas	This study	−136.7+72.1	−136.7+72.1
Alona Bay lavas	This study	−127.3+47.4	−110.0+44.3
Alona Bay lavas	DuBois (1962)	−95.0+39.0	
Mamainse Point lavas	This study	−147.6+54.4	−147.6+54.4
Mean Reverse Polarity Pole, $A_{95} = 13.2°$			−133.5°+50.7°
B. Formations and Magnetic Divisions of Normal Polarity			
Portage Lake lavas	DuBois (1962)	−168.5 26.5	
Portage Lake lavas	Vincenz (1968)	−177.0 24.0	−174.8+25.9
Portage Lake lavas	Books (1968)	−179.0 27.1	
Copper Harbor conglomerate (sediments and lavas)	DuBois (1962)	+176.5 29.0	−183.8+21.0
Copper Harbor (lavas)	Vincenz and Yaskawa (1968)	+176.0 13.0	
Logan Diabase II (dikes)	DuBois (1962)	−171.5 33.0	−171.5+33.0
Duluth Gabbro	DuBois (1962)	−179.5 23.5	−179.5+23.5
Beaver Bay Complex	Beck and Lindsley (1969)	−170.5 27.5	−170.5+27.5
North Shore volcanics	This study	−172.5 32.1	−172.5+32.1
Gargantua Cape volcanics	This study	+179.7 33.9	−180.3+33.9
Michipicoten Island volcanics	This study	+179.0 28.7	−181.0+28.7
Mamainse Point volcanics	This study	−179.6 36.8	−179.6+36.8
Mean Normal Polarity Pole, $A_{95} = 4.3°$			−177.1+29.2°
Mean pole giving unit weight to mean reverse pole and normal pole			−158.9+42.0

yet the estimated ages of these units is similar to the estimated age of the Middle Keweenawan (Obradovich and Peterman 1968; Hunt 1962). This problem was recognized previously by Runcorn (1964).

Relative rotation between the Lake Superior region and the western cordillera of North America is not a likely explanation for the divergence in pole positions (Murthy *et al.* 1968; Palmer 1969). One line of evidence suggests that the Middle Keweenawan pole is younger, but possibly only slightly younger than the poles given by the Belt–Purcell Supergroup. Books (1968, p. 252) reported that normally magnetized flows with declinations south of west underlie the reversely magnetized Ironwood area flows of the "South Trap Range" of Michigan. Although the precision of these results is low, they indicate an equatorial Pacific pole position and may well be the earliest Keweenawan flows in the Lake Superior region.

Should the stratigraphic relationships suggested above be correct, the Middle Keweenawan data from the Lake Superior region indicate a rapid shift in the position of the pole from the equatorial central Pacific to the north central Pacific. Much of the revival in the continental drift hypothesis was brought about by intercontinental comparisons of Phanerozoic polar wander paths. Inferences about Precambrian episodes of drift based upon intercontinental Precambrian apparent polar wander curves are bound to be much less accurate and more tentative if rapid shifts in short periods of time, as evidenced by the Middle Keweenawan and Belt–Purcell results, are a common feature of Precambrian time. This follows as a consequence of the more fragmentary geologic record and of the greater age uncertainties in the dating of Precambrian rock units.

Acknowledgments

The high field demagnetization experiment was completed through the generosity of C. E.

Helsley, who made equipment at the Southwest Center for Advanced Studies, Dallas, available to the author. I thank T. J. Ulrych and N. J. Snelling for the K–Ar analyses of the Logan sill samples, and W. R. VanSchmus for making his Rb/Sr results on the Mamainse Point samples available to me prior to publication. John Starkey made available a file of uniformly distributed directions on a sphere and gave generously of his time in the computer programming for the conglomerate test experiment. R. N. Annels provided a geological map of Michipicoten Island. W. Heinrich, L. Parker, and C. Blackburn made many of the measurements and assisted in the data processing. An early draft of this paper has benefited considerably from the criticism of J. M. Ade-Hall, L. D. Ayres, C. M. Carmichael, P. E. Giblin, and H. C. Halls. The initial field work was done while the author was a recipient of a special fellowship in science from the Canada Council. Later field collections and laboratory work were supported by a grant from the National Research Council of Canada.

ANNELLS, R. N. 1970. Volcanic studies on Michipicoten Island. Geol. Surv. Can., Paper 70–1, Pt. A., pp. 114–117.

AYRES, L. D. 1961. Township 31, Range 19. Ont. Dept. Mines, Prel. Geol. Map No. P. 128.

——— 1969. Geology of Townships 31 and 30, Ranges 20 and 19. Ont. Dept. Mines, Geol. Rept. 69, 100 pp.

BECK, M. E. and LINDSLEY, N. C. 1969. Paleomagnetism of the Beaver Bay Complex, Minnesota. J. Geophys. Res. **74**, pp. 2002–2013.

BOOKS, K. G. 1968. Magnetization of the lowermost Keweenawan lava flows in the Lake Superior area. U.S. Geol. Surv., Prof. Paper 600-D, pp. D245–D254.

BRIDEN, J. C. 1965. Ancient secondary magnetization in rocks. J. Geophys. Res. **70**, pp. 5205–5221.

BURWASH, E. M. 1905. Geology of Michipicoten Island. Univ. Toronto Studies, Geology Series, No. 3.

CREER, K. M. 1962. The dispersion of the geomagnetic field due to secular variation and its determination for remote times from paleomagnetic data. J. Geophys. Res. **67**, pp. 3461–3476.

DUBOIS, P. M. 1962. Paleomagnetism and correlation of Keweenawan rocks. Can. Geol. Surv., Bull. 71, 75 pp.

FAURE, G., CHAUDHURI, S., and FENTON, M. D. 1969. Ages of the Duluth gabbro complex and of the Endion sill, Duluth, Minnesota. J. Geophys. Res. **74**, pp. 720–725.

FISHER, R. 1953. Dispersion on a sphere. Proc. Roy. Soc. London, A, **217**, pp. 295–305.

GIBLIN, P. E. 1969*a*. Kincaid Township. Ont. Dept. Mines, Prel. Geol. Map No. P. 553.

——— 1969*b*. Mamainse Point. Ont. Dept. Mines, Prel. Geol. Map No. P. 554.

——— 1969*c*. Ryan Township. Ont. Dept. Mines, Prel. Geol. Map No. P. 555.

——— 1969*d*. Herrick Township. Ont. Dept. Mines, Prel. Geol. Map No. P. 556.

GIBLIN, P. E. and LEAHY, E. J. 1965. Batchawana sheet. Ont. Dept. Mines, Prel. Geol. Map, No. P. 302.

GOLDICH, S. S., LIDIAK, E. G., HEDGE, C. E., and WALTHALL, F. G. 1966. Geochronology of the midcontinent region, United States. J. Geophys. Res. **71**, pp. 5389–5408.

GOLDICH, S. S., NIER, A. O., BAADSGAARD, H., HOFFMAN, J. H., and KRUEGER, H. W. 1961. The Precambrian geology and geochronology of Minnesota. Minn. Geol. Surv., Bull. 41, 193 pp.

GRAHAM, J. W. 1949. The stability and significance of magnetism in sedimentary rocks. J. Geophys. Res. **54**, pp. 131–167.

——— 1953. Changes of ferromagnetic minerals and their bearing on ferromagnetic properties of rocks. J. Geophys. Res. **58**, pp. 243–260.

GROUT, F. F., SHARP, R. P., and SCHWARTZ, G. M. 1959. The geology of Cook County, Minnesota. Minn. Geol. Surv., Bull. 39, 163 pp.

HALLS, H. C. 1966. A review of the Keweenawan geology of the Lake Superior region. *In* The Earth beneath the Continents. Amer. Geophys. Union, Washington, D.C. Geophys. Monograph 10, pp. 3–27.

HELSLEY, C. E. 1967. Advantages of field drilling of samples for paleomagnetic studies. *In* Methods in Paleomagnetism. Elsevier Publ. Co., New York. pp. 26–30.

HUNT, GRAHAM. 1962. Time of Purcell eruption in southeastern British Columbia and southwestern Alberta. J. Alberta Soc. Petrol. Geol. **10**, pp. 438–442.

IRVING, E. 1964. Paleomagnetism and its application to geological and geophysical problems. John Wiley and Sons Ltd., New York. 339 pp.

——— 1966. Paleomagnetism of some Carboniferous rocks from New South Wales and its relation to geological events. J. Geophys. Res. **71**, pp. 6025–6051.

IRVING, E. and PARRY, L. G. 1963. The magnetism of some Permian rocks from New South Wales. Geophys. J. **7**, pp. 395–411.

LOGAN, W. E. 1863. Geological Survey of Canada, Report of Progress from its commencement to 1863. 983 pp.

LOWDON, J. A., STOCKWELL, C. H., TIPPER, H. W., and WANLESS, R. K. 1963. Age determinations and geological studies. Geol. Surv. Can., Paper 62-17.

MACFARLANE, T. 1866. Report on "Lake Superior". Geol. Surv. Can., Rept. of Progress 1863–66, pp. 132–134.

MCELHINNY, M. W. 1964. Statistical significance of the fold test. Geophys. J. **8**, pp. 338–340.

MURTHY, G. S., FAHRIG, W. F., and JONES, D. L. 1968. The paleomagnetism of the Michikamau anorthosite intrusion, Labrador. Can. J. Earth Sci. **5**, pp. 1139–1144.

NUFFIELD, E. W. 1955. Geology of the Montreal River area. Ont. Dept. Mines, **LXIV**, Pt. 3, pp. 1–32.

OBRADOVICH, J. D. and PETERMAN, Z. E. 1968. Geochronology of the Belt Series, Montana. Can. J. Earth Sci. **5**, pp. 737–747.

PALMER, H. C. 1969. The paleomagnetism of the Croker Island Complex, Ontario, Canada. Can. J. Earth Sci. **6**, pp. 213–218.

RUNCORN, S. K. 1964. Paleomagnetic results from Precambrian sedimentary rocks in the western United States. Bull. Geol. Soc. Amer. **75**, pp. 687–704.

SANDBERG, A. E. 1938. Section across Keweenawan lavas at Duluth, Minnesota. Bull. Geol. Soc. Amer. **49**, pp. 795–830.

SILVER, L. T. and GREEN, J. C. 1963. Zircon ages for Middle Keweenawan rocks of the Lake Superior region (Abstr.). Trans. Amer. Geophys. Union, **44**, p. 107.

TANTON, T. L. 1931. Fort William and Port Arthur, and Thunder Cape map-areas, Thunder Bay District, Ontario. Geol. Surv. Can., Memoir **167**, 222 pp.

THOMSON, JAS. E. 1953. Geology of the Mamainse Point Copper area. Ont. Dept. Mines, **LXII**, Pt. 4, pp. 1–25.

VINCENZ, S. A. 1968. Phenomenon of partial self-reversal in Keweenawan rocks. 1, Magnetization of Portage Lake lavas. J. Geophys. Res. **73**, pp. 2729–2752.

VINCENZ, S. A. and YASKAWA, K. 1968. Phenomenon of partial self-reversal in Keweenawan rocks. 2, Magnetization of Upper Keweenawan lavas and sediments and of Lower Keweenawan dikes. J. Geophys. Res. **73**, pp. 2753–2767.

WILSON, R. L. and HAGGERTY, S. E. 1966. Reversals of the earth's magnetic field. Endeavour, **25**, pp. 104–109.